BioVision AI Platform: User Manual

Version 1.0.0

1. Overview

The BioVision AI Platform is an integrated deep learning suite designed to bridge the gap between complex AI architectures and practical biomedical research. It provides a user-friendly Graphical User Interface (GUI) to perform two primary tasks:

- **Instance Segmentation:** A hybrid YOLOv11 and SAM 2.1 workflow for precise boundary detection.
- **Image Classification:** A high-speed YOLOv11 engine for category-level identification.

2. Prerequisites & Installation

Before using the software, ensure your environment is configured according to the **Guidelines for BioVision installation software.pdf** found on the BioVision GitHub Repository.

3. Module 1: Instance Segmentation

The Instance Segmentation module utilizes a "detect-and-segment" approach. YOLOv11 identifies the object and provides a bounding box, which acts as an automated prompt for SAM 2.1 to generate a high-fidelity mask.

Operating Instructions

- **Tab Selection:** Click the **"Instance Segmentation"** tab in the top navigation bar.
- **Configuration:**
    - Click **Select YOLO Model** and choose your trained .pt weights.
    - Click **Select SAM Model** and choose a SAM 2.1 checkpoint (Base, Large, Small, or Tiny).

- Click **Select Image Folder** to load your dataset (JPG, PNG, or TIFF).
- Click **Select Output Directory** to choose where results will be stored.

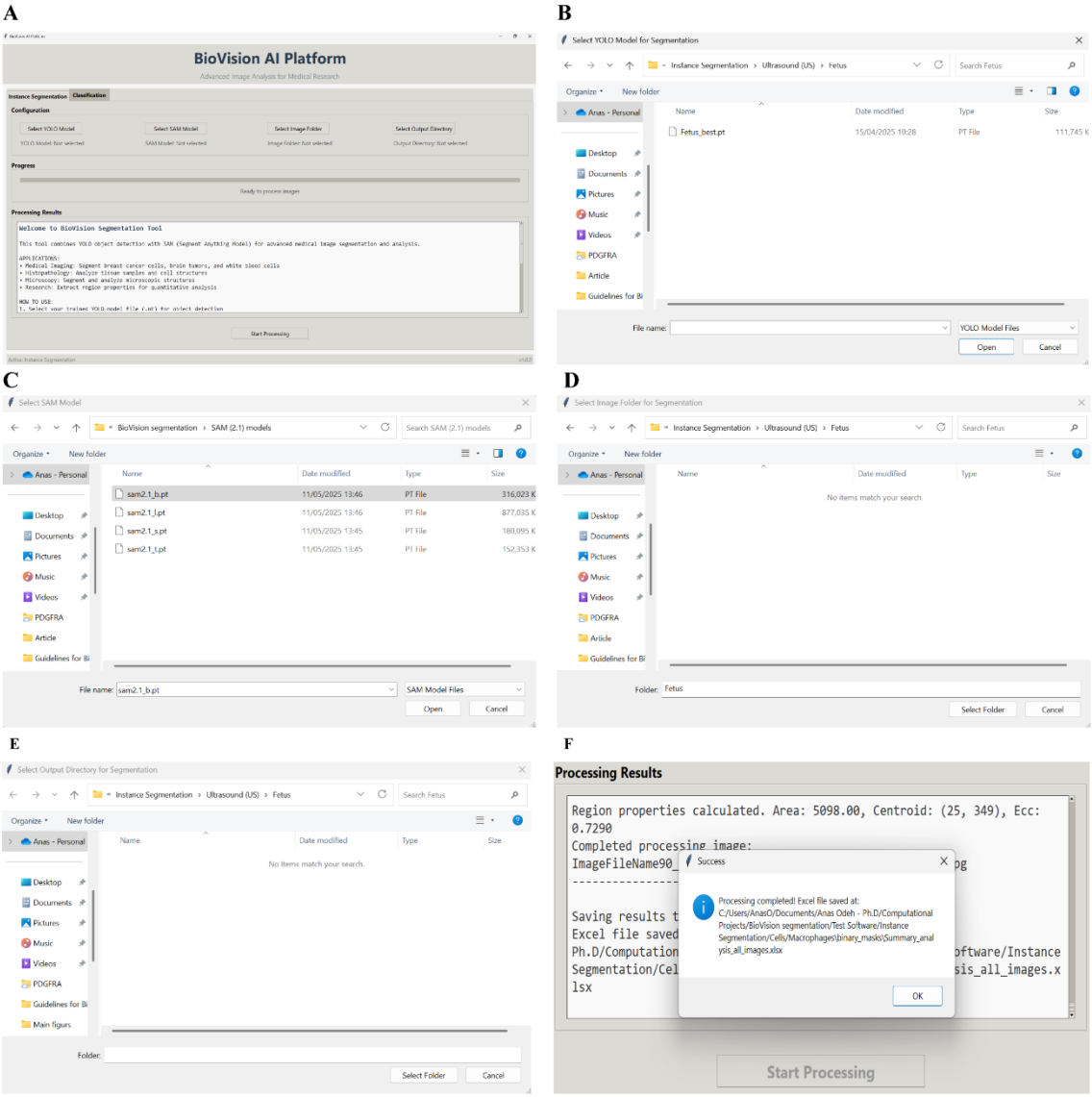

Fig. 1. BioVision Instance Segmentation Module: User Interface and Configuration Workflow.

(A) The primary Instance Segmentation interface, featuring the modular configuration panel for model initialization and directory setup. **(B)** Model selection dialog for loading fine-tuned YOLOv11 weights (e.g., Fetus_best.pt) to serve as the region proposal engine. **(C)** Selection interface for the SAM 2.1 transformer-based mask decoder, supporting multiple architecture variants (Base, Large, Small, Tiny). **(D)** Input directory configuration for designating the source folder containing standardized medical or histological imagery for batch processing.

- **Execution:** Click **Start Processing**. The progress bar will indicate the real-time status of the batch analysis.
- **Results:** Once complete, the software saves individual masks and a summary report to your designated output folder.

4. Module 2: Image Classification

This module is optimized for high-throughput classification tasks, such as distinguishing between healthy and pathological tissues or subtyping cells.

Operating Instructions

- **Tab Selection:** Click the **"Classification"** tab.
- **Model Selection:** Click **Select YOLO Model** to choose your specialized classification weights.
- **Directory Setup:**
    - Use the **Select Image Folder** button to load your batch of images.
    - Use the **Select Output Directory** button to set the destination for your reports.

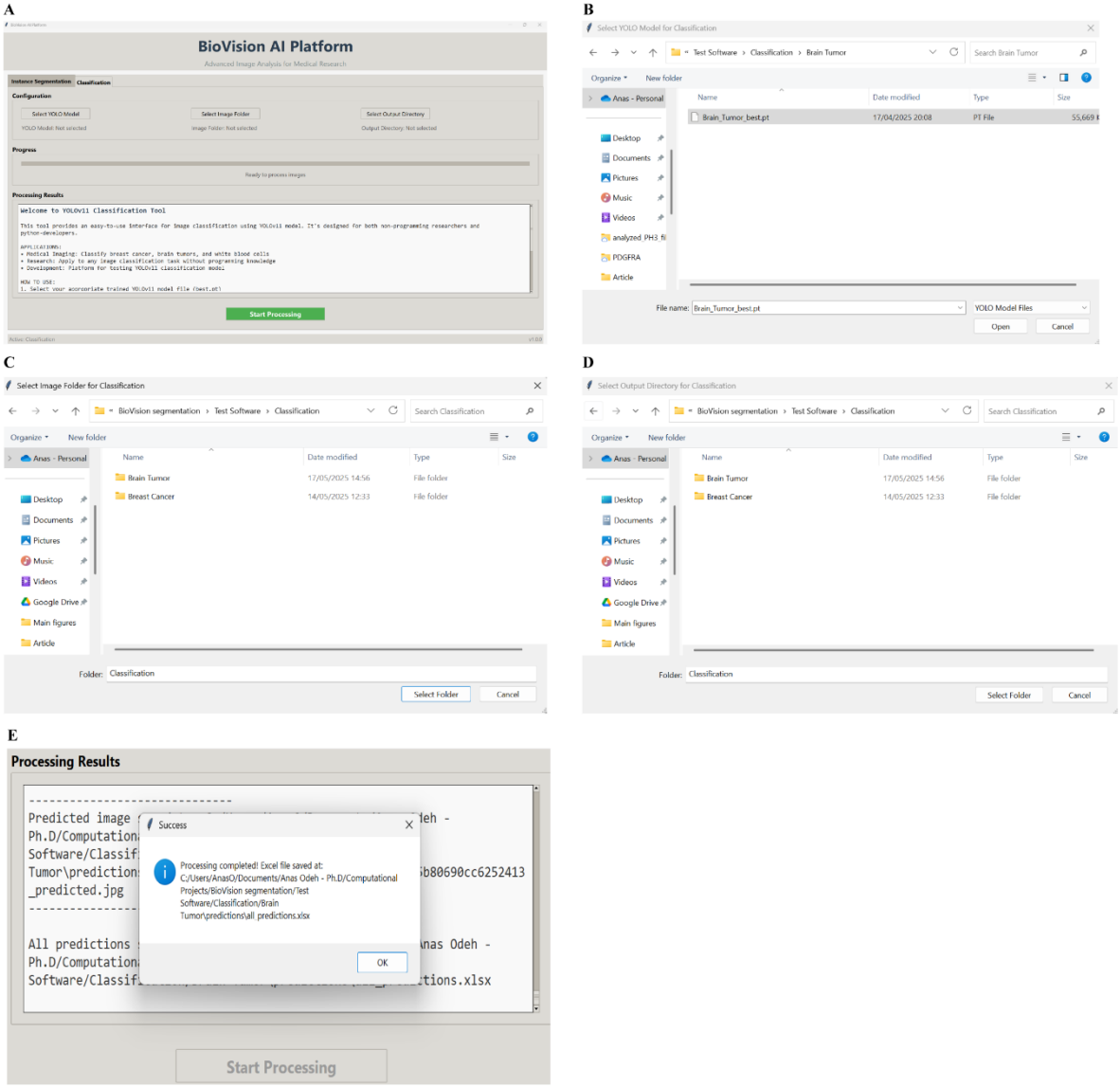

Fig. 2. BioVision Classification Module: Batch Analysis and Results Export.

(A) The Classification tab of the BioVision AI Platform, optimized for high-throughput diagnostic labeling. **(B)** Selection of specialized YOLO-based classification models (e.g., Brain_Tumor_best.pt) for targeted clinical tasks. **(C)** Input directory selection window for organizing batch datasets. **(D)** Output directory configuration interface for directing processed results. **(E)** The post-processing success dialog and results log, confirming the automated generation of comprehensive Excel (.xlsx) reports containing prediction classes and associated confidence scores.

- **Execution:** Click **Start Processing**.
- **Report Retrieval:** A success dialog will appear confirming the completion. The software automatically generates an **Excel (.xlsx)** file containing:
    - Image Name
    - Predicted Class
    - Confidence Score

5. Data Export & Technical Support

All visual predictions are translated into structured, machine-readable formats (CSV/Excel) to facilitate downstream statistical analysis.

- **For Source Code:** https://github.com/Anas-Odeh/BioVision
- **For Pre-trained Models:** Access the Google Drive links provided in the main manuscript.
- **Technical Support:** For bug reports or feature requests, please open an "Issue" on the GitHub repository.